

Bluestock Mutual Fund Analytics

1. Executive Summary

The Bluestock Mutual Fund Analytics project is a complete data analytics and business intelligence solution designed to study mutual fund industry performance, investor activity, fund-house assets under management, SIP inflows, category-wise flows, portfolio information, and benchmark movements.

The project processes ten structured CSV datasets containing fund master information, daily NAV history, fund-house AUM, monthly SIP inflows, category-level inflows, folio counts, scheme performance, investor transactions, portfolio holdings, and benchmark indices.

The main objective is to convert raw financial datasets into meaningful analytical outputs that can support fund comparison, risk analysis, investor segmentation, industry monitoring, and management decision-making.

The project includes:

- Data ingestion and cleaning
- Exploratory data analysis
- SQLite database creation
- Fund performance calculations
- Risk-adjusted ranking
- Investor transaction analysis
- SIP and category-flow analysis
- Interactive Streamlit dashboard
- Exported dashboard screenshots
- Final PDF report and presentation

The dataset contains approximately 40 mutual fund schemes and 32,778 investor transactions. The data period mainly covers 2022 to 2025.

The analysis measures return, volatility, Sharpe ratio, beta, maximum drawdown, historical Value at Risk, AUM, SIP inflows, category inflows, transaction value, and investor demographics.

The project is intended for analytical and educational purposes. The recommendations generated by the system are decision-support outputs and should not be considered financial advice.

2. Project Objectives

The major objectives of the project are:

1. To create a reliable ETL pipeline for mutual fund data.
2. To clean and standardize raw CSV files.
3. To create a structured SQLite database.
4. To study industry-level AUM and folio trends.
5. To compare mutual fund returns and risk.
6. To calculate risk-adjusted performance metrics.
7. To analyze investor transactions by state, age group, city tier, and transaction type.
8. To study SIP inflows and category-wise net inflows.
9. To compare mutual fund NAV performance with the NIFTY 50 benchmark.
10. To create an interactive dashboard for business users.
11. To generate a reproducible PDF report and presentation.
12. To design an explainable fund-ranking algorithm.

3. Data Sources

The project uses ten major datasets.

3.1 Fund Master Dataset

This dataset contains scheme-level information such as:

- AMFI code
- Scheme name
- Fund house
- Category
- Sub-category
- Plan type
- Benchmark
- Launch date
- Risk grade
- Rating

The fund master table acts as the primary reference table for scheme-level analysis.

3.2 NAV History Dataset

The NAV history table contains daily NAV observations for each mutual fund scheme.

Important columns include:

- AMFI code
- Date
- NAV

This dataset is used to calculate daily returns, annualized returns, volatility, drawdown, and Value at Risk.

3.3 Fund-House AUM Dataset

This table records assets under management by fund house over time.

Important fields include:

- Date
- Fund house
- AUM in lakh crore

This dataset is used to analyse industry scale and AMC concentration.

3.4 Monthly SIP Inflow Dataset

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It helps identify:

- Growth in SIP participation
- Monthly investment patterns
- Relationship between SIP flows and market levels
- Increasing or decreasing investor confidence

3.5 Category Inflow Dataset

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Examples of categories may include:

- Equity
- Debt
- Hybrid
- Large Cap
- Mid Cap
- Small Cap

- Flexi Cap
- Index Funds

3.6 Folio Count Dataset

3.6 Folio Count Dataset

It indicates the number of investor accounts and helps measure retail participation.

3.7 Scheme Performance Dataset

3.7 Scheme Performance Dataset

Common fields include:

- One-year return
- Standard deviation
- Sharpe ratio
- Beta
- AUM
- Morningstar rating
- Risk grade
- Fund category

3.8 Investor Transactions Dataset

3.8 Investor Transactions Dataset

Important fields include:

- Transaction date
- State
- Age group
- City tier
- Transaction type
- Transaction amount

Transaction types may include:

- SIP
- Lumpsum
- Redemption

3.9 Portfolio Holdings Dataset

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Possible fields include:

- Scheme name
- Security name
- Sector
- Instrument type
- Portfolio weight

This dataset can be used to analyse sector exposure and portfolio concentration.

3.10 Benchmark Dataset

3.10 Benchmark Dataset

It is used to calculate:

- Benchmark returns
- Fund beta
- Fund-versus-index comparisons
- Relative performance

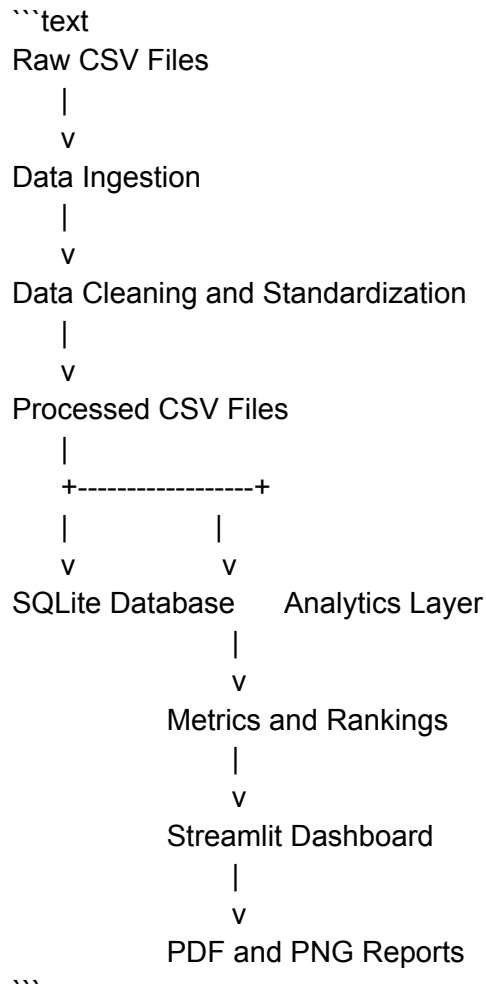
4. Technology Stack

The project uses the following technologies:

Component	Technology
---	---
Programming Language	Python
Data Processing	Pandas, NumPy
Visualization	Plotly, Matplotlib
Dashboard	Streamlit
Database	SQLite
Report Generation	Matplotlib PdfPages
Presentation	python-pptx
Data Storage	CSV
Development Environment	Visual Studio Code
Notebook Analysis	Jupyter Notebook

5. Overall System Architecture

The project follows a layered architecture.



The raw datasets are preserved separately from the cleaned datasets. This makes it possible to reproduce the entire workflow and compare raw records with processed outputs.

6. ETL Design

ETL stands for:

- Extract
- Transform
- Load

6.1 Extract

The extraction stage reads the ten raw CSV files from the `raw` directory.

The pipeline checks whether each required input file exists before processing.

If a required file is missing, the pipeline raises an error instead of silently continuing.

6.2 Transform

The transformation stage performs the following operations:

1. Reads the CSV file.
2. Removes leading and trailing spaces from column names.
3. Converts column names to lowercase.
4. Parses date columns.
5. Removes duplicate rows.
6. Replaces infinite numeric values with missing values.
7. Resets the row index.
8. Saves the cleaned dataset.

Date fields include:

- `date`
- `month`
- `transaction\_date`
- `launch\_date`

6.3 Load

The cleaned datasets are saved into the `processed` directory.

The processed files are then loaded into SQLite tables. The database provides a structured environment for SQL queries and analytical reporting.

7. ETL Cleaning Logic

The core cleaning function is:

```
```python
def clean_frame(frame):
```

```

frame = frame.copy()

frame.columns = [
 column.strip().lower()
 for column in frame.columns
]

for column in DATE_COLUMNS.intersection(frame.columns):
 frame[column] = pd.to_datetime(
 frame[column],
 errors="coerce"
)

frame = frame.drop_duplicates()
frame = frame.reset_index(drop=True)

numeric_columns = frame.select_dtypes(
 include="number"
).columns

frame[numeric_columns] = frame[numeric_columns].replace(
 [float("inf"), float("-inf")],
 pd.NA
)

return frame
...

```

This logic ensures that every input file follows a consistent structure.

The use of `errors="coerce"` prevents invalid date values from stopping the entire pipeline. Invalid dates are converted to missing values and can later be examined through data-quality checks.

## 8. Database Design

The cleaned tables are stored in SQLite.

The database provides the following advantages:

- Structured storage
- SQL querying

- Reusable analytical tables
- Easier data validation
- Faster filtering and aggregation
- Separation between raw data and reporting outputs

Typical tables include:

- `fund\_master`
- `nav\_history`
- `aum\_by\_fund\_house`
- `monthly\_sip\_inflows`
- `category\_inflows`
- `industry\_folio\_count`
- `scheme\_performance`
- `investor\_transactions`
- `portfolio\_holdings`
- `benchmark\_indices`

Indexes can be created on frequently used fields such as:

- AMFI code
- Date
- Transaction date
- Fund house
- Category

## 9. Exploratory Data Analysis

Exploratory Data Analysis, or EDA, was performed to understand the structure, distribution, trends, and relationships within the data.

The following questions were considered:

- Which fund houses have the highest AUM?
- How does industry AUM change over time?
- Which categories receive the highest net inflows?
- How do SIP inflows move over time?
- Which states contribute the highest transaction value?
- Which investor age groups transact more?
- What is the distribution of transaction types?
- Which funds provide higher returns at lower risk?
- How do selected funds compare with the NIFTY 50?

## 10. Industry-Level Findings

The industry analysis focuses on AUM, fund-house concentration, SIP inflows, and folio growth.

The AUM chart aggregates assets by date and fund house. This makes it possible to identify the largest fund houses and observe changes in industry scale.

The fund-house ranking chart highlights the top AMCs based on the latest available AUM.

AUM should not be interpreted as a direct measure of investment quality. A fund house may have high AUM because of its age, distribution network, brand recognition, or large institutional accounts.

SIP inflows measure new recurring contributions, while AUM measures the total value of assets. These two indicators should be analysed separately.

For example:

- Increasing AUM with increasing SIP inflows may indicate strong growth.
- Increasing AUM with declining SIP inflows may be caused by market appreciation.
- Stable AUM with high redemptions may indicate investor withdrawals.
- Increasing folios with stable AUM may indicate smaller average investments.

## 11. Investor Analytics

Investor transactions were analysed using the following dimensions:

- State
- Age group
- City tier
- Transaction type
- Transaction date
- Transaction amount

The transaction-mix analysis compares SIP, lumpsum, and redemption amounts.

SIP transactions indicate recurring investment behaviour. Lumpsum transactions represent one-time investments. Redemptions represent withdrawals from mutual fund investments.

State-level analysis helps identify geographic concentration.

Age-group analysis helps compare average transaction amounts across investor cohorts.

City-tier analysis can provide information about participation from:

- Tier 1 cities
- Tier 2 cities
- Tier 3 cities

Monthly transaction volume helps identify periods of higher or lower investor activity.

However, transaction value does not measure investment profitability. A high transaction amount may represent either a purchase or a redemption, so transaction type must always be considered.

## 12. SIP and Category-Flow Analysis

SIP inflows were compared with NIFTY 50 levels using a dual-axis chart.

The chart contains:

- SIP inflow on the primary axis
- NIFTY 50 closing value on the secondary axis

This comparison helps identify whether SIP contributions remain stable during changing market conditions.

Category-level inflows were visualized using a heatmap.

The heatmap makes it easier to identify:

- Categories with persistent inflows
- Categories with repeated outflows
- Seasonal flow patterns
- Strong and weak months
- Differences between equity and non-equity categories

A separate chart displays the top categories by March or financial-year-end inflow.

Category inflow is not the same as fund return. A category may receive large inflows even when recent performance is weak because of investor preference, tax planning, product launches, or market expectations.

### 13. Mutual Fund Performance Analysis

The performance analysis uses daily NAV data.

For a fund with NAV values  $NAV_t$  and  $NAV_{t-1}$ , the daily return is calculated as:

$$R_t = \frac{NAV_t}{NAV_{t-1}} - 1$$

The daily returns are then used to calculate performance and risk metrics.

#### 13.1 Annualized Return

The total return is calculated as:

$$R_{\text{total}} = \frac{NAV_{\text{final}}}{NAV_{\text{initial}}} - 1$$

The annualized return is:

$$R_{\text{annual}} = (1 + R_{\text{total}})^{1/Y} - 1$$

where:

$$Y = \frac{\text{Number of daily observations}}{252}$$

The value 252 represents the approximate number of trading days in one year.

#### 13.2 Annualized Volatility

Annualized volatility is calculated as:

\$\$

$$\sigma_{\text{annual}} = \sigma_{\text{daily}} \times \sqrt{252}$$

\$\$

A higher volatility indicates greater variation in daily returns.

### 13.3 Sharpe Ratio

The Sharpe ratio measures return relative to risk.

The project assumes a 6% annual risk-free rate.

The daily risk-free rate is:

\$\$

$$R_{f,\text{daily}} = \frac{0.06}{252}$$

\$\$

The Sharpe ratio is:

\$\$

Sharpe =

$$\frac{\overline{R_d} - R_{f,\text{daily}}}{\sigma_d}$$

$\times \sqrt{252}$

$$\times \sqrt{252}$$

\$\$

A higher Sharpe ratio generally indicates better risk-adjusted performance.

### 13.4 Beta

Beta measures the sensitivity of fund returns to benchmark returns.

\$\$

$\beta =$

$$\frac{\text{Cov}(R_{\text{fund}}, R_{\text{benchmark}})}{\text{Var}(R_{\text{benchmark}})}$$

$$\frac{\text{Cov}(R_{\text{fund}}, R_{\text{benchmark}})}{\text{Var}(R_{\text{benchmark}})}$$

\$\$

Interpretation:

- Beta greater than 1: more sensitive than the benchmark
- Beta near 1: similar sensitivity to the benchmark
- Beta below 1: less sensitive than the benchmark

- Negative beta: moves in the opposite direction

### 13.5 Maximum Drawdown

Cumulative wealth is calculated as:

$$W_t = \prod_{i=1}^t (1 + R_i)$$

The running maximum is:

$$M_t = \max(W_1, W_2, \dots, W_t)$$

Drawdown is:

$$D_t = \frac{W_t}{M_t} - 1$$

Maximum drawdown is the minimum value of  $D_t$ .

It measures the largest decline from a previous peak.

### 13.6 Historical Value at Risk

Historical 95% daily VaR is calculated as the fifth percentile of daily returns:

$$VaR_{95} = \text{Percentile}(R_{\text{daily}}, 5\%)$$

This represents a historical estimate of a severe daily loss threshold.

VaR is not a guarantee. It depends entirely on the observed return history.

## 14. Risk-Metric Algorithm

The main algorithm can be summarized as:

```
```python
for fund_code, fund_data in nav.groupby("amfi_code"):
```

```
    prices = fund_data.sort_values("date")
    prices = prices.set_index("date")["nav"]
```

```
    daily_returns = prices.pct_change().dropna()
```

```
    years = len(daily_returns) / 252
```

```
    total_return = (
        prices.iloc[-1] / prices.iloc[0]
    ) - 1
```

```
    annual_return = (
        (1 + total_return) ** (1 / years)
    ) - 1
```

```
    volatility = (
        daily_returns.std()
        * np.sqrt(252)
    )
```

```
    excess_returns = (
        daily_returns - 0.06 / 252
    )
```

```
    sharpe_ratio = (
        excess_returns.mean()
        / daily_returns.std()
    ) * np.sqrt(252)
```

```
    cumulative_returns = (
        1 + daily_returns
    ).cumprod()
```

```
    drawdown = (
        cumulative_returns
        / cumulative_returns.cummax()
    ) - 1
```

```
    maximum_drawdown = drawdown.min()
```

```
    var_95 = daily_returns.quantile(0.05)
```

...

Each fund is processed separately so that scheme-level risk statistics can be compared.

15. Fund Recommendation Algorithm

The recommendation system uses an explainable scoring model.

The algorithm supports optional filters for:

- Fund category
- Risk grade

After filtering, percentile ranks are calculated for:

- Sharpe ratio
- One-year return
- Annualized volatility

The score is:

```
$$
Score =
0.50 \times Rank(Sharpe)
+
0.30 \times Rank(Return)
+
0.20 \times (1 - Rank(Volatility))
$$
```

The score gives the highest weight to risk-adjusted performance.

The recommendation logic is:

```
```python
view["recommendation_score"] = (
 view["sharpe_ratio"].rank(pct=True) * 0.50
 + view["return_1yr_pct"].rank(pct=True) * 0.30
 + (
 1 - view["std_dev_ann_pct"].rank(pct=True)
) * 0.20
)
```

```
result = view.sort_values(
 ["recommendation_score", "sharpe_ratio"],
 ascending=False
).head(limit)
'''
```

This model is transparent because every score can be explained using the three input components.

The model does not consider:

- Investor income
- Investment horizon
- Tax status
- Liquidity requirements
- Existing portfolio
- Financial goals
- Market forecasts
- Personal risk capacity

Therefore, the recommendation should be treated as an analytical ranking and not as investment advice.

## 16. Dashboard Design

The project dashboard is developed using Streamlit.

It contains four pages.

### 16.1 Industry Overview

This page displays:

- Total AUM
- SIP inflows
- Folio count
- Number of schemes
- Industry AUM trend
- AUM by fund house

Users can filter by:

- Fund house
- AUM date

## 16.2 Fund Performance

This page displays:

- Return versus risk scatter plot
- Fund-level risk classification
- NAV versus NIFTY 50 chart
- Scheme scorecard
- Fund-house filter
- Category filter
- Plan filter

The NAV comparison chart rebases both fund NAV and benchmark value to 100.

The rebasing formula is:

$$\text{NormalizedValue}_t = \frac{\text{Value}_t}{\text{Value}_{\text{initial}}} \times 100$$

This allows two series with different absolute values to be compared visually.

## 16.3 Investor Analytics

This page displays:

- Transaction amount by state
- Transaction-type mix
- Average amount by age group
- Monthly transaction volume

It supports filtering by:

- State
- Age group
- City tier

## 16.4 SIP and Market Trends

This page displays:

- Monthly SIP inflows
- NIFTY 50 levels
- Category inflow heatmap
- Top five categories by financial-year inflow

The dashboard is intended to help users explore the data interactively rather than relying only on static tables.

## 17. Dashboard Screenshots

The report includes the following dashboard screenshots:

1. Industry Overview
2. Fund Performance
3. Investor Analytics
4. SIP and Market Trends

Each screenshot demonstrates the implemented analytical functionality and confirms that the dashboard is connected to the processed data layer.

The screenshots should be placed in the report with captions such as:

- Figure 1: Industry Overview Dashboard
- Figure 2: Fund Performance Dashboard
- Figure 3: Investor Analytics Dashboard
- Figure 4: SIP and Market Trends Dashboard

## 18. Limitations

The project has several limitations.

### 18.1 Historical Data Limitation

The analysis is based on historical observations. Historical performance does not guarantee future performance.

### 18.2 Data Completeness

Some datasets may be incomplete or simulated. Missing values, limited history, or inconsistent source coverage may affect results.

#### ### 18.3 Survivorship Bias

##### 18.3 Survivorship Bias

#### ### 18.4 Benchmark Alignment

##### 18.4 Benchmark Alignment

#### ### 18.5 Risk-Free Rate Assumption

##### 18.5 Risk-Free Rate Assumption

#### ### 18.6 Historical VaR Limitation

##### 18.6 Historical VaR Limitation

#### ### 18.7 No Investor Suitability Model

##### 18.7 No Investor Suitability Model

#### ### 18.8 No Live Production Refresh

##### 18.8 No Live Production Refresh

#### ### 18.9 Power BI Limitation

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### ## 19. Recommendations

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##### 19.1 Improve Data Quality

Add automated checks for:

- Missing columns
- Unexpected data types

- Duplicate records
- Null values
- Invalid dates
- Negative NAV values
- Missing benchmark observations
- Duplicate AMFI codes

## 19.2 Add Scheduled Refresh

The pipeline can be scheduled using:

- Windows Task Scheduler
- Apache Airflow
- Azure Data Factory
- GitHub Actions
- Cloud-based job scheduling

## 19.3 Add Advanced Metrics

Future versions may include:

- Rolling one-year return
- Rolling volatility
- Alpha
- Information ratio
- Sortino ratio
- Tracking error
- Upside capture
- Downside capture
- Calmar ratio
- Expense ratio analysis

## ### 19.4 Improve Recommendation Logic

### 19.4 Improve Recommendation Logic

- Minimum observation requirements
- Category-specific benchmarks
- Investor risk profile
- Investment horizon
- Fund expense ratio
- Exit load
- Portfolio concentration
- Downside-risk weighting

## 19.5 Add Portfolio Analysis

The holdings data can be used to calculate:

- Sector concentration
- Top-ten holding concentration
- Equity/debt allocation
- Portfolio turnover
- Common holdings across schemes
- Portfolio overlap between funds

## ### 19.6 Improve Governance

### 19.6 Improve Governance

- Input file versions
- Processing timestamps
- Metric assumptions
- Data-quality results
- Dashboard export dates
- Recommendation model versions

## 20. Reproducibility

The project can be reproduced from the repository root using:

```
```powershell
python scripts/run_pipeline.py
```
```

The pipeline performs the following steps:

1. Reads raw CSV files.
2. Cleans and standardizes the data.
3. Writes processed CSV files.
4. Loads data into SQLite.
5. Calculates fund metrics.
6. Exports dashboard assets.
7. Generates the PDF report.
8. Generates the presentation.

The project structure is organized as follows:

```
``text
data/raw/
 Original source CSV files

data/processed/
 Cleaned analysis-ready CSV files

scripts/
 ETL and analytics scripts

src/
 Database and query utilities

notebooks/
 Exploratory and analytical notebooks

database/
 SQLite database and exports

reports/
 Final PDF report and presentation

app.py
 Streamlit dashboard application
...
```

## 21. Conclusion

The Bluestock Mutual Fund Analytics project demonstrates how raw mutual fund data can be transformed into a complete analytical product.

The project combines data engineering, financial analytics, risk measurement, investor segmentation, visualization, database design, and dashboard development.

The ETL pipeline creates a repeatable foundation for analysis. The risk metrics provide a structured method for comparing returns and risk. The investor analysis reveals transaction behaviour across geography, age group, and city tier. The SIP and category-flow analysis provides an industry-level view of investment activity.

The Streamlit dashboard makes the results accessible to business users through interactive filters and charts.

The recommendation engine provides an explainable ranking based on Sharpe ratio, return, and volatility. However, the output must be interpreted carefully because it does not replace professional financial advice or investor suitability analysis.

Overall, the project provides a strong foundation for a production-grade mutual fund analytics platform. Future improvements should focus on live data ingestion, stronger validation, advanced portfolio analytics, investor suitability modelling, and automated reporting.